

PSC XI ENTRY

SOFTWARE / EDGE-AI

PROTOTYPE STAGE

TingínHB

”See Anemia Before It Kills.”

Non-invasive, dual-site edge-AI hemoglobin screening designed for frontline Philippine community healthcare. Zero consumables, zero needles, 100% offline on budget smartphones in under two seconds.

► TEAM DATALUNAS · USTP College of IT & Computing

TEAM IDENTITY

DataLunas (USTP)

CORE INNOVATORS

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ACADEMIC BASE

Dept. of Data Science — USTP

ECOSYSTEM TARGET

75,000+ Barangay Health Workers (BHWs)

DICT IIDB 2026

16:9 KEYNOTE

The Philippine Anemia Crisis & Diagnostic Desert

Postpartum Hemorrhage (PPH) is the #1 obstetric killer in the country. Anemia robs mothers of the physiological reserve to survive delivery, yet community diagnostic coverage is practically zero.

21.8%

Pregnant Filipinas entering labor clinically anemic
(DOST-FNRI 2024)

2,000+

Preventable maternal deaths/yr from Postpartum Hemorrhage (PPH)

40–45%

Filipino infants (6–11 mos) facing irreversible 5–10 IQ point loss

>50%

Anemic cases completely missed by naked-eye visual pallor examination

🚨 The Pathophysiology Triad

Anemic mothers ($\text{Hb} < 9.0 \text{ g/dL}$) face **2.5× to 4.0× higher fatal delivery risk**. Myometrial smooth muscle lacks ATP to contract (causing uterine atony in 70–80% of PPH). A normal mother tolerates 500mL blood loss; an anemic mother decompensates into irreversible shock after just 300mL.

🏠 The Primary Care Diagnostic Gap

Over **200,000 BHWs** carry maternal screening on their shoulders with no tools. **HemoCue Hb 301** units cost ₱70k–125k + **₱150 per disposable strip**, creating chronic stockouts. Subjective physical inspection has poor reliability ($\kappa = 0.20 \text{--} 0.45$).

Direct SDG Priorities & Philippine Statutory Mandates

TingínHB is wired into national healthcare legislation and UN sustainable development benchmarks, providing an actionable technical mechanism for institutional policy targets.

3.1 Maternal MMR

Target MMR <70 per 100k live births. PH currently struggles at **119–144**. TingínHB directly attacks the primary cause of PPH fatalities.

3.2 Child Survival

End preventable neonatal and infant deaths. Early microvascular screening protects the critical First 1,000 Days cognitive window.

3.8 Universal UHC

Achieve essential health service coverage. Software-only distribution eliminates geographic and procurement barriers in remote GIDAs.

10.3 Equal Access

Eradicate diagnostic disparity. Equips rural BHWs with identical objective screening precision enjoyed by tertiary private hospitals.

📄 RA 11148 (First 1,000 Days)

Mandates routine maternal & child nutritional screening across barangays. TingínHB delivers ₱0–cost compliance for home visits.

⚖️ RA 11223 (Universal Health Care)

Expands comprehensive primary care benefit packages in Geographically Isolated and Disadvantaged Areas (GIDAs).

🏠 PhilHealth Konsulta Integration

Acts as the first-line community triage filter, generating standardized referral slips for accredited CBC laboratory verification.

Target Beneficiaries & Market Sizing

A three-tier operational ecosystem: frontline BHWs deploy the tool, vulnerable families receive life-saving triage, and government healthcare systems capture millions in consumable cost-avoidance.

Primary Operators (Users)

~75,000 **Barangay Health Workers (BHWs)** and Rural Health Unit (RHU) midwives conducting antenatal outreach and routine Well-Baby consultations.

Direct Beneficiaries (Patients Screened)

1.7M **pregnant women/year** entering antenatal care, 11M **children under 5**, and 15–20% of **adolescent schoolgirls** enrolled in DepEd weekly IFAS programs.

Institutional Buyers (Customers)

DOH Central & Regional Offices, Local Government Units (LGUs) with UHC Special Health Funds, and maternal health NGOs (UNFPA, Helen Keller Intl).

Quantifiable Market Opportunity

Total Addressable Market (TAM)	900,000 screenings/mo
National Consumable Avoidance	₱135,000,000 / month
Regional Beachhead (Mindanao SOM)	10 LGUs (~18k tests/mo)
Software SaaS Potential (@10% BHWs)	₱3.75M Annual ARR

Geographic Beachhead

Pilot deployments launching across Northern Mindanao GIDA barangays, leveraging USTP's academic footprint where physician density routinely drops below 1 doctor per 20,000 residents.

The TingínHB Multi-Site Solution

Point, focus, and capture. TingínHB digitizes physician pallor inspection by evaluating two complementary microvascular sites, executing 100% on-device in under 2 seconds on a budget ₱5,000 Android smartphone.

👁️ Primary Site: Palpebral Conjunctiva

The mucosal lining of the lower eyelid contains **zero melanin pigment**, guaranteeing uniform optical baseline performance across all Filipino skin complexions (Fitzpatrick III–VI). Adjacent sclera provides an in-frame white balance reference.

👉 Fallback Site: Subungual Fingernail Bed

Non-intrusive alternative when eyelid eversion is impractical (distressed infants 6–24 mos, active conjunctivitis, photophobia). Differential Erythema Index (ΔEI) cancels periungual melanin bias.

⚡ Inverse-Variance Fusion Engine

When both sites are photographed, the system calculates an uncertainty-weighted composite reading. Discrepancies exceeding 2.0 g/dL trigger an immediate rescreen flag for clinical safety.

TIER	HB RANGE	ACTION PROTOCOL
GREEN	$\text{Hb} \geq 11.0 \text{ g/dL}$	Normal Baseline. Continue standard ANC & Iron-Folic Acid.
YELLOW	$10.0 \text{--} 10.9 \text{ g/dL}$	Mild Anemia. Review adherence; dietary counseling; follow-up.
ORANGE	$7.0 \text{--} 9.9 \text{ g/dL}$	Moderate Anemia. Immediate RHU referral for confirmatory CBC.
RED	$\text{Hb} < 7.0 \text{ g/dL}$	Severe Anemia. URGENT Hospital Referral — High PPH Death Risk.

📄 One-Tap PhilHealth Konsulta Referral

For Orange/Red cases, TingínHB auto-compiles an encrypted PDF referral containing patient metadata, gestational age, MUAC, and screening confidence intervals. Ready for direct clinical submission with zero manual paperwork.

Six Core Engineering Features

Designed for an entry-level MediaTek Helio G35 or Qualcomm Snapdragon 400 processor in an off-grid barangay station, not a temperature-controlled tertiary hospital laboratory.

13.3 MB INT8 Edge AI

100% on-device quantization. Dual YOLOv8-nano-seg models + MobileNetV3 small regression heads execute in <60ms per site with zero cloud sync.

Real-Time Quality Gate

Laplacian blur thresholding ($\text{Var} > 120$) and exposure bounding reject out-of-focus or glare-ruined photos before inference occurs.

Melanin-Agnostic Normalization

In-frame sclera color correction ($k_c = 255 / \overline{\text{Sclera}_c}$) and differential fingernail contrast eliminate external calibration cards.

4-Tier WHO Clinical Triage

Instant color-coded risk stratification mapped to DOH AO 2010-0010 micronutrient directives with visual dual-site confidence bars.

Automated PDF Referral

Instant generation of standardized PhilHealth Konsulta referral forms ready for Bluetooth printing or WhatsApp/SMS relay.

Offline Encrypted SQLite

Zero-leakage local storage preserves longitudinal maternal hemoglobin trajectories across monthly checkups without cellular connectivity.

Freemium Public Health Model & Moat

BHWs have zero purchasing power; monetizing end users breaks primary care access. Revenue flows from institutional cost avoidance, LGU software procurement, and public health surveillance analytics.

DOH & LGU Annual SaaS Licensing (Year 3)

Municipal health offices procure TingínHB at **₱500 / BHW / year**. Covering 10% of national BHWs yields **₱3.75M ARR**.

Maternal Health NGO Deployments (Year 2)

Programmatic software licenses for UNFPA, PATH, and Helen Keller International at **₱50,000–₱150,000 / program**.

100% Free Core Screening for Frontline BHWs

The mobile screener remains permanently free for all certified BHWs and community midwives to maximize nationwide adoption.

DIMENSION	TINGÍNHB	HEMOCUE 301	ANEMOCHECK (US)
Cost / Screening	₱0.00	₱105–₱150	~\$5 + baseline CBC
Hardware Cost	₱0 (Phone)	₱70k–₱125k	Smartphone only
Anatomical Sites	Eye + Nail	Fingerstick	Nail Bed Only
Prior Lab CBC	NOT Required	No	Mandatory Initial CBC
Melanin Independence	ScLera Ref.	Yes	Melanin Sensitive
Offline Edge Execution	100% Local	Yes	Yes

Current Traction & 18-Month Strategic Roadmap

From clinical literature audit to a functioning on-device Flutter prototype. We are advancing through structured academic validation toward clinical field trials in Northern Mindanao.

✓ Completed Milestones

- 1. Clinical Burden Audit:** Full descriptive synthesis of DOST-FNRI, WHO, and DOH maternal mortality data.
- 2. Dataset Curation:** Audited 6 global benchmark datasets (CP-AnemiC, EYES-DEFY, Mannino) totaling 1,200+ subjects.
- 3. Optical Preprocessing Pipeline:** Mathematical formulation of in-frame sclera color correction and ΔEI .
- 4. Flutter Mobile Prototype:** CameraX live reticle UI, tri-mode selector, and PDF referral compiler completed.

➡ Strategic Next Steps (Post-PSC)

- Months 1–4:** INT8 TFLite model optimization & on-device benchmarking across low-tier Android SoCs.
- Months 5–10:** Northern Mindanao Clinical Trial (≥ 100 pregnant mothers paired with laboratory CBC ground truth).
- Months 11–14:** PFDA Class B SaMD regulatory pre-assessment filing under ASEAN Medical Device Directive.
- Months 15–18:** DICT / DOH pilot deployment across 10 GIDA municipalities with LGU sponsorship.

₱280,000 Seed Requirement & Triple Ask

A lean, one-time development capitalization to validate the core neural architecture against local clinical ground truth and initiate Philippine FDA software clearance.

BUDGET	ALLOCATION ITEM	AMOUNT (PHP)	DELIVERABLE
	Clinical Validation Study (\$n \ge 100\$)	₱80,000	Paired hospital CBC ground truth & IRB compliance
	GPU Training Compute & Cloud	₱80,000	Multi-site segmentation & quantization pipeline
	Regulatory & PFDA SaMD Consultation	₱30,000	Class B classification filing under ASEAN AMDD
	Mobile Device Test Matrix (3 tiers)	₱30,000	Cross-device camera sensor calibration
	Mobile App & PDF Tools	₱20,000	Production Flutter deployment build
	BHW Field Manuals & Training	₱15,000	Video & printed training curriculum
	Contingency Buffer (10%)	₱25,000	Field trial logistics & supplies
	TOTAL SEED REQUIREMENT	₱280,000	12-Month Field Validation

💰 1. Seed Grant Capital

Targeted via PSC XI Award Funds, DICT Startup Grant Fund (SGF), and USTP institutional research support.

🏥 2. Regional Clinical Partner

Introductions to Northern Mindanao RHUs, district hospitals, or maternity clinics for paired non-invasive & CBC blood validation.

🧭 3. HealthTech & Regulatory Mentorship

Guidance from DOH Digital Health experts and FDA medical device software regulatory practitioners.

THE THESIS

See Anemia. Save Mothers.

By converting 200,000 existing BHW smartphones into zero-cost clinical screening sensors, we eliminate the diagnostic blind spot that claims 2,000 Filipino mothers every year.

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Philippine Startup Challenge XI · DICT IIDB 2026

 ₱0 Cost Per Test

Eliminates ₱150 microcuvette barrier across 75,000 frontline health workers.

 <2s Edge-AI Triage

Instant WHO risk grading and automated PhilHealth Konsulta referral generation.

 2,000 Lives Protected

Detects severe anemia before fatal uterine atony triggers irreversible delivery shock.

Notice: TinginHB is a Class B Clinical Decision Support & Triage Aid under the ASEAN AMDD. Not a replacement for definitive venous CBC.